Design Analysis and Algorithm – Lab Work Week 5

WEEK - 7

JOB SEQUENCING

CODE:

```c
#include <stdio.h>
struct Job {
    char id;
    int dead, profit;
};

int main() {
    int n, maxD = 0, totalP = 0;
    printf("Number of jobs: ");
    scanf("%d", &n);
    struct Job arr[n];

    for(int i=0; i<n; i++) {
        printf("Job %d (ID Deadline Profit): ", i+1);
        // Added space before %c to prevent the double-prompt bug
        scanf(" %c %d %d", &arr[i].id, &arr[i].dead, &arr[i].profit);
        if(arr[i].dead > maxD) maxD = arr[i].dead;
    }

    for(int i=0; i<n-1; i++) {
        for(int j=0; j<n-i-1; j++) {
            if(arr[j].profit < arr[j+1].profit) {
                struct Job tmp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = tmp;
            }
        }
    }

    int slot[maxD];
    char res[maxD];
    for(int i=0; i<maxD; i++) slot[i] = 0;

    for(int i=0; i<n; i++) {
        for(int j = (arr[i].dead < maxD ? arr[i].dead : maxD) - 1; j >= 0; j--) {
            if(slot[j] == 0) {
                res[j] = arr[i].id;
                slot[j] = 1;
                totalP += arr[i].profit;
                break;
            }
        }
    }

    printf("\nSequence: ");
    for(int i=0; i<maxD; i++) if(slot[i]) printf("%c ", res[i]);
    printf("\nTotal Profit: %d\n", totalP);
    return 0;
}
```

OUTPUT:

```
Number of jobs: 14
Job 1 (ID Deadline Profit): A 3 22
Job 2 (ID Deadline Profit): B 3 19
Job 3 (ID Deadline Profit): C 8 29
Job 4 (ID Deadline Profit): D 6 28
Job 5 (ID Deadline Profit): E 7 30
Job 6 (ID Deadline Profit): F 5 21
Job 7 (ID Deadline Profit): G 10 27
Job 8 (ID Deadline Profit): H 4 25
Job 9 (ID Deadline Profit): I 6 24
Job 10 (ID Deadline Profit): J 12 26
Job 11 (ID Deadline Profit): K 13 14
Job 12 (ID Deadline Profit): L 2 27
Job 13 (ID Deadline Profit): M 14 19
Job 14 (ID Deadline Profit): N 1 11

Sequence: F L A H I D E C G J K M
Total Profit: 292
```

TIME COMPLEXITY: uses sorting and scheduling and also finding max of numbers i.e. (n^2),n,nxd= O(n^2 + n + n x d =O(n^2).

SPACE COMPLEXITY: O(n + d).

SOLVED:

CH.SC.U4CSE24137

JOB SEQUENCING

1. jobs = 14

profits = {22, 19, 29, 28, 30, 21, 27, 25, 24, 26, 14, 27, 19, 11}

deadlines = {3, 3, 8, 6, 7, 5, 10, 4, 6, 12, 13, 8, 14, 13}

job = {J5, J3, J4, J7, J12, J10, J8, J9, J1, J6, J2, J13, J11, J14}

profit = {30, 29, 28, 27, 27, 26, 25, 24, 22, 21, 19, 19, 14, 11}.

deadlines = {7, 8, 6, 10, 2, 12, 4, 6, 3, 5, 3, 14, 13, 13}

step 1:

→

J6	J12	J1	J8	J9	J4	J5	J3		J7		J10	J11	J13

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

1) 30, 7 (J5), P = 30

2) 29, 8 (J3), P = 59

3) 28, 6 (J4), P = 87

4) 27, 10 (J7), P = 114

5) 27, 2 (J12), P = 141

6) 26, 4 (J10), P = 167

7) 25, 4 (J8), P = 192

8) 24, 6 (J9), P = 216
cannot be added but J5 is empty so

9) 22, 3 (J1), P = 236

10) 21, 5 (J6), P = 259

5 X
4 X
3 X
2 X
1 ✓

11) 19, 3 (J2), P = X

9 X
8 X
1 X, so skipped

12) 19, 14 (J13)

13) 14, 13 (J11), P = 292

14) 11, 1 (J14), P = X
1 - X

final ⇒

J6	J12	J1	J8	J9	J4	J5	J3		J7		J10	J11	J13

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

max profit = 292.